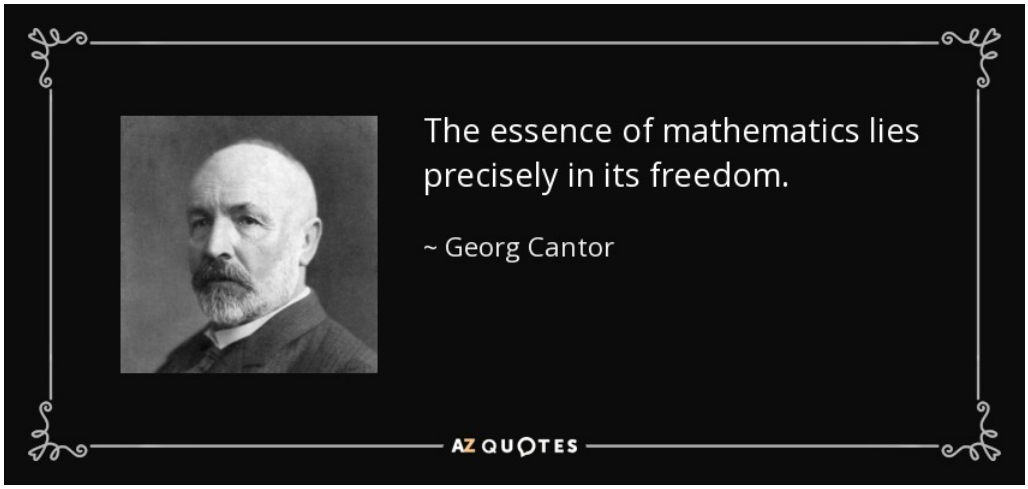
Chapter 1 - Logic and Proof

Note 3

Lecturer: Dr. Rajitha Ranasinghe Academic Year: 2024/2025 - Semester: II



1.3 : Techniques of Proof : I

Our aim is to be able to read and write mathematics, and this requires more than just vocabulary. It also requires syntax. That is, we need to understand how statements are combined to form the mysterious mathematical entity known as a proof. Let us first look at the two main types of logical reasoning: ***inductive reasoning*** and ***deductive reasoning***.

Example 1 : Consider $f(n) = n^2+n+17$. If we evaluate this for various positive integers, we observe that we always seem to obtain a prime number. For example,

$$f(1) = 19$$
$$f(2) = 23$$
$$f(3) = 29$$
$$\vdots$$
$$f(8) = 89$$
$$\vdots$$
$$f(12) = 173$$
$$\vdots$$
$$f(15) = 257$$

and all these numbers (as well as the ones skipped over) are prime. On the basis of this experience we might conjecture that the function $f(n) = n^2 + n + 17$ will always produce a prime number when $n \in \mathbb{Z}^+$. Drawing a conclusion of this sort is an example of **inductive reasoning**. On the basis of looking at individual cases we make a general conclusion.

If we let $p(n)$:"$n^2 + n + 17$ is a prime number" for $n \in \mathbb{Z}^+$, then we can ask, is $\forall\ n,\ p(n)$ a true statement ? Have we proved it is true ?

It is important to realize that we have not proved that it is true. We have shown that $\exists\ n \ni\ p(n)$ is true. In fact, we know that $p(n)$ is true for many n. But we have not proved that it is true for all n. It turns out that we cannot, since the statement "$\forall\ n,\ p(n)$" happens to be false.

How do we know that it is false? Because we can think of an example where $n^2 + n + 17$ is not prime. (Such an example is called a **counterexample**.) One such counterexample is $n = 17$:

$$17^2 + 17 + 17 = 17(17 + 1 + 1) = 17 \cdot 19.$$

There are others as well. For example, when $n = 16$,

$$\begin{aligned} 16^2 + 16 + 17 &= 16(16 + 1) + 17 \\ &= 16(17) + 17 = (16 + 1)(17) = 17^2 \end{aligned}$$

but it only takes one counterexample to prove that " $\forall\ n,\ p(n)$ " is false.

On the basis of Example 1 we might infer that inductive reasoning is of little value. Although it is true that the conclusions drawn from inductive reasoning have not been proved logically, they can be very useful. Indeed, this type of reasoning is the basis for most if not all scientific experimentation.

Example 2 : Provide counterexamples to the following statements.

(a) All birds can fly.

(b) Every continuous function is differentiable.

Solution:

(a) Any flightless bird, such as an ostrich.

(b) The absolute value function is continuous for all real numbers, but it is not differentiable at the origin.

Example 3: Consider the function $g(n, m) = n^2 + n + m$, where n and m are understood to be positive integers. In Example 1 we saw that $g(16, 17) = 16^2 + 16 + 17 = 17^2$. We might also observe that

$$g(1, 2) = 1^2 + 1 + 2 = 4 = 2^2$$
$$g(2, 3) = 2^2 + 2 + 3 = 9 = 3^2$$
$$\vdots$$
$$g(5, 6) = 5^2 + 5 + 6 = 36 = 6^2$$
$$\vdots$$
$$g(12, 13) = 12^2 + 12 + 13 = 169 = 13^2$$

On the basis of these examples (using inductive reasoning) we can form the conjecture "$\forall\ n,\ q(n)$," where $q(n)$ is the statement $g(n, n + 1) = (n + 1)^2$. It turns out that our conjecture this time is true, and we can prove it. Using the familiar laws of algebra, we have

$$\begin{aligned} g(n, n+1) &= n^2 + n + (n+1) && [\text{definition of} g(n, n+1)] \\ &= n^2 + 2n + 1 && [\text{ since } n + n = 2n] \\ &= (n+1)(n+1) && [\text{ by factoring }] \\ &= (n+1)^2 && \left[\text{ definition of } (n+1)^2\right]. \end{aligned}$$

Since our reasoning at each step does not depend on n being any specific integer, we conclude that " $\forall\ n,\ q(n)$ " is true.

Now that we have proved the general statement " $\forall\ n,\ q(n)$," we can apply it to any particular case. Thus, we know that $g(124, 125) = 125^2$ without having to do any computation. This is an example of **deductive reasoning**: applying a general principle to a particular case. Most of the proofs encountered in mathematics are based on this type of reasoning.

Example 4: In what way was deductive reasoning used in Example 3 to prove $\forall\ n,\ q(n)$?

Solution: The general rules about factoring polynomials were applied to the specific polynomial $n^2 + n + (n + 1)$.

The most common type of mathematical theorem can be symbolized as $p \Longrightarrow q$, where p and q may be compound statements. To assert that $p \Longrightarrow q$ is a theorem is to claim that $p \Longrightarrow q$ is a tautology; that is, that it is always true. From Note 1 we know that $p \Longrightarrow q$ is true unless p is true and q is false.

Thus, to prove that p implies q, we have to show that whenever p is true it follows that q must be true. When an implication $p \Longrightarrow q$ is identified as a theorem, we refer to p as the **hypothesis** and q as the **conclusion**.

The construction of a proof of the implication $p \Longrightarrow q$ can be thought of as building a bridge of logical statements to connect the hypothesis p with the conclusion q. The building blocks that go into the bridge consist of four kinds of statements:

1. Definitions
2. Assumptions or axioms that are accepted as true
3. Theorems that have previously been established as true
4. Statements that are logically implied by the earlier statements in the proof

The logical equivalences discussed in Note 1 provide alternate ways to join the blocks together. When building the bridge, it may not be at all obvious which blocks to use or in what order to use them. This is where experience is helpful, together with perseverance, intuition, and sometimes a good bit of luck.

We hope that at some point the part of the bridge leaving p will join with the part that arrives at q, forming a complete span:

$$p \Longrightarrow r_1 \Longrightarrow r_2 \Longrightarrow \cdots \Longrightarrow r_{n-1} \Longrightarrow r_n \Longrightarrow q.$$

Example 5 : Let us return to the result proved in Example 3 to illustrate the process just described. We begin by writing the theorem in the form $p \Longrightarrow q$. One way of doing this is as follows: "If $g(n, m) = n^2 + n + m$, then $g(n, n + 1) = (n + 1)^2$." Symbolically, we identify the hypothesis

$$p : g(n, m) = n^2 + n + m$$

and the conclusion

$$q : g(n, n + 1) = (n + 1)^2.$$

In asking what statement will imply q, there are many possible answers. One such answer is by taking the following:

$$r_1 : g(n, n + 1) = n^2 + n + (n + 1)$$

$$r_2 : g(n, n + 1) = n^2 + 2n + 1$$

$$r_3 : g(n, n + 1) = (n + 1)(n + 1)$$

it follows that $p \Longrightarrow r_1 \Longrightarrow r_2 \Longrightarrow r_3 \Rightarrow q$. This is essentially what was written in Example 3.

Associated with an implication $p \Longrightarrow q$ there is a related implication $\sim q \Longrightarrow \sim p$, called the **contrapositive**. Using a truth table, it is easy to see that an implication and its contrapositive are logically equivalent. Thus one way of proving an implication is to prove its contrapositive.

Example 6 :

(a) Use a truth table to verify that $p \Longrightarrow q$ and $\sim q \Longrightarrow \sim p$ are logically equivalent.

(b) Is $p \Longrightarrow q$ logically equivalent to $q \Longrightarrow p$?

Solution :

(a)

p	q	$(p \Rightarrow q)$	$\Leftrightarrow$	$[(\sim q) \Rightarrow (\sim p)]$
T	T	T	T	F T F
T	F	F	T	T F F
F	T	T	T	F T T
F	F	T	T	T T T

(b) No, $p \Rightarrow q$ is not logically equivalent to $q \Rightarrow p$. The biconditional between them is not a tautology.

p	q	$(p \Rightarrow q)$	$\Leftrightarrow$	$(q \Rightarrow p)$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	F
F	F	T	T	T

Example 7 : The contrapositive of the theorem "If $7m$ is an odd number, then m is an odd number" is "If m is not an odd number, then $7m$ is not an odd number" or, equivalently, "If m is an even number, then $7m$ is an even number." Using the contrapositive, we can construct a proof of the theorem as follows:

Hypothesis: m is an even number.

- Then we can write $m = 2k$ for $k \in \mathbb{Z}$.
- It follows that $7m = 2(7k)$.

Conclusion: $7m$ is an even number.

This is much easier than trying to show directly that $7m$ being odd implies that m is odd.

Example 8 : Write the contrapositive of each implication in Example 5 in Note 1.

Solution :

(a). If $2n$ is an odd number, then n is not an integer.

(b). If you do not have a college degree, then you cannot work here.

(c). If the car runs, then you are not out of gas.

(d). If a function is not continuous, then it is not differentiable.

In Example 6 (b) we saw that $p \Longrightarrow q$ is not logically equivalent to $q \Longrightarrow p$. The implication $q \Longrightarrow p$ is called the **converse** of $p \Longrightarrow q$. It is possible for an implication to be false, while its converse is true. Thus we cannot prove $p \Longrightarrow q$ by showing $q \Longrightarrow p$.

Example 9 : The implication "If $m^2 > 0$, then $m > 0$ " is false, but its converse "If $m > 0$, then $m^2 > 0$ " is true.

Example 10 : Write the converse of each implication in Example 5 in Note 1.

Solution :

(a) If $2n$ is an even number, then n is an integer.

(b) If you have a college degree, then you can work here.

(c) If the car does not run, then you are out of gas.

(d) If a function is continuous, then it is differentiable.

Another implication that is closely related to $p \Longrightarrow q$ is the **inverse** $\sim p \Longrightarrow \sim q$. The inverse implication is not logically equivalent to $p \Longrightarrow q$, but it is logically equivalent to the converse. The inverse is the contrapositive of the converse.

Example 11 : Use a truth table to show that the inverse and the converse of $p \Longrightarrow q$ are logically equivalent.

Solution:

p	q	$(q \Rightarrow p)$	$\Leftrightarrow$	$[(\sim p) \Rightarrow (\sim q)]$
T	T	T	T	F T F
T	F	T	T	F T T
F	T	F	T	T F F
F	F	T	T	T T T

Looking at the contrapositive form of an implication is a useful tool in proving theorems. Since it is a property of the logical structure and does not depend on the subject matter, it can be used in any setting involving an implication. There are many more tautologies that can be used in the same way. Some of the more common ones are listed in the next example.

Example 12 : The following tautologies are useful in constructing proofs. The first two indicate, for example, that an "if and only if" theorem $p \Longleftrightarrow q$ can be proved by establishing $p \Rightarrow q$ and its converse $q \Longrightarrow p$ or by showing $p \Longrightarrow q$ and its inverse $\sim p \Longrightarrow \sim q$. The letter c is used to represent a statement that is always false. Such a statement is called a **contradiction**. While this list of tautologies need not be memorized, it will be helpful if each one is studied carefully to see just what it is saying. In the next section we illustrate the use of several of these tautologies.

(a). $(p \Longleftrightarrow q) \Longleftrightarrow [(p \Longrightarrow q) \wedge (q \Longrightarrow p)]$

(b). $(p \Longleftrightarrow q) \Longleftrightarrow [(p \Longrightarrow q) \wedge (\sim p \Longrightarrow \sim q)]$

(c). $(p \Longrightarrow q) \Longleftrightarrow (\sim q \Longrightarrow \sim p)$

(d). $p \vee \sim p$

(e). $(p \wedge \sim p) \Longleftrightarrow c$

(f). $(\sim p \Longrightarrow c) \Longleftrightarrow p$

(g). $(p \Longrightarrow q) \Longleftrightarrow [(p \wedge \sim q) \Longrightarrow c]$

(h). $[p \wedge (p \Longrightarrow q)] \Longrightarrow q$

(i). $[(p \Longrightarrow q) \wedge \sim q] \Longrightarrow \sim p$

(j). $[(p \vee q) \wedge \sim p] \Longrightarrow q$

(k). $(p \wedge q) \Longrightarrow p$

(l). $[(p \Longrightarrow q) \wedge (q \Longrightarrow r)] \Longrightarrow (p \Longrightarrow r)$

(m). $[(p_1 \Longrightarrow p_2) \wedge (p_2 \Longrightarrow p_3) \wedge \cdots \wedge (p_{n-1} \Longrightarrow p_n)] \Longrightarrow (p_1 \Longrightarrow p_n)$

(n). $[(p \wedge q) \Longrightarrow r] \Longleftrightarrow [p \Longrightarrow (q \Longrightarrow r)]$

(o). $[(p \Longrightarrow r) \wedge (r \Longrightarrow s) \wedge (p \vee r)] \Longrightarrow (q \vee s)$

(p). $[p \Longrightarrow (q \vee r)] \Longleftrightarrow [(p \wedge \sim q) \Longrightarrow r]$

(q). $[(p \vee q) \Longrightarrow r] \Longleftrightarrow [(p \Longrightarrow r) \wedge (q \Longrightarrow r)]$

.....